Elyes Darouich

Mobile Software Engineer

5th Year Engineering Student at ESPRIT

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Check my Portfolio
23 years old

Professional Summary

Passionate mobile software engineer in 5th year at ESPRIT (3ème année cycle d'ingénieur), specializing in cross-platform mobile development. Expert in iOS (SwiftUI, UIKit), Android (Kotlin), Flutter, and full-stack web development with React, NestJS, Next.js, and Spring Boot. Experienced in AI/ML integration, DevOps practices, and agile methodologies with proven track record in enterprise-level projects and innovative mobile applications.

Technical Skills

Programming Languages

Mobile & Native: Java, Kotlin, Swift, Dart/Flutter, Objective-C

Web & Backend: JavaScript, TypeScript, PHP, Python, C++, C

Database: SQL, NoSQL, MongoDB, MySQL, Firebase, SQLite

Mobile Development

iOS Development: SwiftUI, UIKit, Core Data, MapKit, iOS SDK, Xcode

Android Development: Kotlin, Java, Android SDK, Jetpack Compose, Android Studio

Cross-Platform: Flutter, Dart, React Native, Xamarin

Web & Backend Technologies

Frontend: React, Next.js, Vue.js, HTML5, CSS3, Bootstrap, Tailwind CSS

Backend: NestJS, Node.js, Spring Boot, .NET, Symfony, Express.js

APIs: REST, GraphQL, WebSockets, Socket.IO, Microservices

Cloud & DevOps

Cloud Platforms: AWS, Google Cloud, Firebase, Vercel

DevOps: Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins

Tools: Git/GitHub, Linux, PowerBI, Arduino, Postman

AI & Machine Learning

AI Integration: Google Gemini API, OpenAI GPT, TensorFlow Lite

ML Concepts: Natural Language Processing, Computer Vision, Deep Learning

Methodologies & Practices

Development: Agile/Scrum, Test-Driven Development, Clean Architecture

Design: UML, Figma, Adobe XD, Responsive Design, UI/UX Principles

Professional Experience

Engineering Intern - Mobile & Backend Development — **Dec 2024 - Feb 2025**
Talan Tunisia — *Tunis, Tunisia*

- Designed and developed **ProjectFlow** - a comprehensive cross-platform project management platform using Flutter & NestJS architecture
- Implemented robust authentication system with JWT + Google OAuth integration and role-based access control

- Built responsive Kanban board with drag-and-drop functionality across mobile, web, and desktop platforms
- Integrated real-time collaboration features using WebSockets and Socket.IO for seamless team communication
- Developed AI Assistant powered by Google Gemini 2.0 Flash for intelligent task automation and project insights
- Implemented comprehensive DevOps pipeline with Docker containerization and GitHub Actions
- Achieved exceptional performance with API response times ¡150ms and stable performance for 100+ concurrent users
- Contributed to 78% test coverage with comprehensive unit and integration testing using Jest and Flutter Test
- Sprint management and analytics dashboard implementation with real-time data visualization

Flutter Developer Intern — **Summer 2023**
Vatech WABAG — *Tunis, Tunisia*

- Developed comprehensive Flutter application for automating employee leave management processes
- Built and integrated RESTful API for seamless backend communication and real-time data processing
- Designed intuitive user interface with real-time push notifications and status updates
- Collaborated extensively with HR team to gather functional requirements and implement business logic
- Implemented secure authentication system with role-based access control for different user types
- Optimized app performance and user experience across Android and iOS platforms

Industrial Trainee — **Summer 2022**
Vatech WABAG — *Tunis, Tunisia*

- Gained comprehensive understanding of company management and organizational processes
- Developed strong teamwork and internal communication skills through cross-functional collaboration
- Learned complete IT project lifecycle in enterprise environment from planning to deployment
- Participated actively in cross-functional team meetings and strategic planning sessions

Key Projects

ProjectFlow - AI Project Management Platform — **2024-2025**

- **Technologies:** Flutter, NestJS, MongoDB, Docker, Google Gemini AI, WebSockets, JWT, OAuth
- Cross-platform project management solution with intelligent AI integration for task automation
- Real-time Kanban boards with drag-and-drop functionality across mobile, web, and desktop platforms
- AI assistant for automated task creation, deadline suggestions, and project performance insights
- WebSocket-powered real-time collaboration with instant updates and team communication features
- Comprehensive user management with role-based access control and Google OAuth integration
- Advanced analytics dashboard with sprint management and performance tracking

Dialex - Language Learning Platform — **2024-2025**

- **Technologies:** SwiftUI, Kotlin, NestJS, MongoDB, iOS SDK, Android SDK
- Cross-platform language learning application for iOS and Android with native implementations
- Interactive exercises and quiz system with gamification features and progress tracking
- Voice recognition integration for pronunciation practice and feedback
- RESTful API backend built with NestJS and MongoDB for user data and learning content management
- Point-based reward system with leaderboards and achievement tracking
- Personalized learning paths based on user performance and preferences

Employee Leave Management System — **2023**

- **Technologies:** Flutter, REST API, Firebase, Push Notifications, Material Design
- Comprehensive leave management automation system for enterprise-level deployment
- Automated leave request and approval workflow with multi-level authorization
- Real-time notifications and status updates for all stakeholders
- Integration with HR systems and comprehensive reporting dashboard
- Role-based access control with different interfaces for employees, managers, and HR

E-Commerce Platform 2023

- **Technologies:** React, Next.js, Node.js, MySQL, Stripe API, AWS S3
- Full-stack e-commerce solution with modern UI/UX and secure payment processing
- Advanced product catalog with search, filtering, and recommendation engine
- Secure payment gateway integration with Stripe for multiple payment methods
- Admin dashboard for inventory management, order processing, and sales analytics
- Responsive design optimized for desktop, tablet, and mobile devices

University Management System 2022

- **Technologies:** Java, Spring Boot, MySQL, Thymeleaf, Bootstrap
- Comprehensive academic management system for student and faculty administration
- Student enrollment, course management, and grade tracking functionality
- Faculty dashboard for course creation, assignment management, and student evaluation
- Administrative tools for semester planning, room scheduling, and academic reporting

Content Management Website 2022

- **Technologies:** PHP, Symfony, MySQL, HTML/CSS, JavaScript, Bootstrap
- Dynamic content management system with custom admin interface
- Responsive design optimized for all devices with modern UI components
- User-friendly admin panel for content creation, editing, and publishing
- Database integration with optimized queries for fast content retrieval

2D Game Development with SDL 2021

- **Technologies:** C, SDL Library, Game Physics, Custom Graphics Engine
- Custom 2D video game with advanced game mechanics and collision detection system
- Comprehensive scoring system with level progression and difficulty scaling
- Custom level design tools and asset creation pipeline
- Optimized game physics engine for smooth character movement and interactions

Education

Computer Engineering - Mobile Development Specialization — **2021 – Present**
ESPRIT (École Supérieure Privée d'Ingénierie et de Technologies) — *Ariana, Tunisia*

- Currently in 5th year (3ème année cycle d'ingénieur) with specialization in mobile application development
- Completed 4th year engineering internship at Talan Tunisia developing ProjectFlow platform
- Advanced coursework in software architecture, AI integration, system design, and enterprise development
- Hands-on experience with modern development frameworks, DevOps practices, and agile methodologies
- Active participation in hackathons, programming competitions, and technical workshops
- Strong academic foundation in algorithms, data structures, databases, and software engineering principles
- Cross-platform development expertise gained through comprehensive mobile development curriculum

Baccalauréat in Computer Science — **2021**
Lycée Jeune Fille — *Tunisia*

- Graduated with specialization in Computer Science and advanced Mathematics
- Strong foundation in programming fundamentals, logical thinking, and problem-solving methodologies
- Active participation in national mathematics and computer science competitions
- Early development of passion for software development and innovative technology solutions

Languages

Arabic: Native Speaker

French: Advanced (B2)

English: Fluent (C1)

Certificates

AWS Academy Graduate - Cloud Foundations — **2025**

- Completed comprehensive AWS Cloud Foundations training program
- Earned AWS Academy Graduate Training Badge demonstrating cloud computing proficiency
- Gained expertise in AWS core services, cloud architecture, and security best practices
- Practical experience with EC2, S3, RDS, VPC, and other essential AWS services

Hashgraph Developer Course — **2025**

- Successfully completed 100% of the Hashgraph Developer Course curriculum
- Comprehensive training in distributed ledger technology and Hedera Hashgraph platform
- Hands-on experience with smart contracts, consensus algorithms, and decentralized applications
- Practical knowledge of building blockchain solutions using Hashgraph technology

Generative AI with Diffusion Models 2025

- Certificate of Competency in Generative AI with Diffusion Models
- Advanced understanding of diffusion models for generative AI applications
- Practical experience with image generation, text-to-image synthesis, and AI model training
- Application of cutting-edge AI techniques in creative and technical projects

Achievements

Successfully completed engineering internship at Talan Tunisia with excellent performance review

Participated in multiple hackathons and programming competitions at national and university level

Contributed to open-source projects on GitHub with focus on mobile development and AI integration

Self-taught in advanced AI/ML technologies including Google Gemini API and OpenAI integration

Mentored junior students in mobile development and software engineering best practices

Active member of ESPRIT programming clubs and technical communities

Interests & Activities

Technical Interests: Problem Solving, AI & Machine Learning, Mobile Innovation

Personal Interests: Gaming, Technology Trends, Continuous Learning

Professional Development: Hackathons, Programming Competitions, Open Source

Community: Mentoring, Technical Workshops, Student Organizations